

Class Size Trend Effect on Student Performance in Early Childhood Education

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Background - Project STAR

- Large-scale, longitudinal, randomized experiment on the effect of class size on student performance in their early childhood education.
- Tracked more than 6,000 students each year in 79 Tennessee schools between 1985 and 1989.

Background - Established Research

- Finn, 1998
 - Small classes **significantly benefit** students in the **early grades**
 - Benefits are **strongest for at-risk students**
 - Small classes improve classroom engagement and behavior
- Finn et al., 2001
 - Students with 3 - 4 years of exposure showed the **strongest long-term achievement gains**.
- Schanzenbach, 2007
 - Smaller classes in the early elementary grades lead to statistically significant improvements in student achievement
 - Students who begin in small classes in kindergarten **and remain in them for multiple years** experience the largest gains

Research Goals

The surrounding literature suggests that smaller class sizes lead to statistically significantly better grades. We aim to address the following related questions:

1. Do smaller class sizes disproportionately benefit low-income students in early elementary school?
2. For students who begin their academic journey in regularly sized classrooms, is it educationally advantageous to switch to smaller classes in later years?*

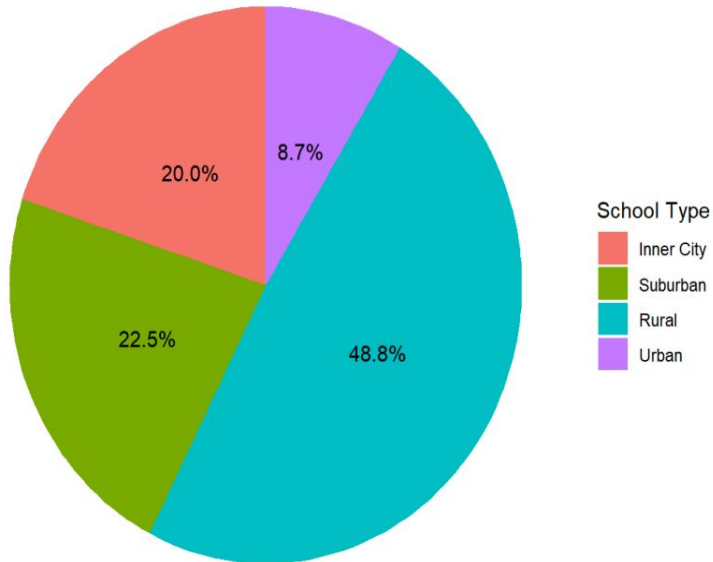
In our case, **student success** is measured by performance on the end-of-year **Math SAT's**.

Why do these questions matter?

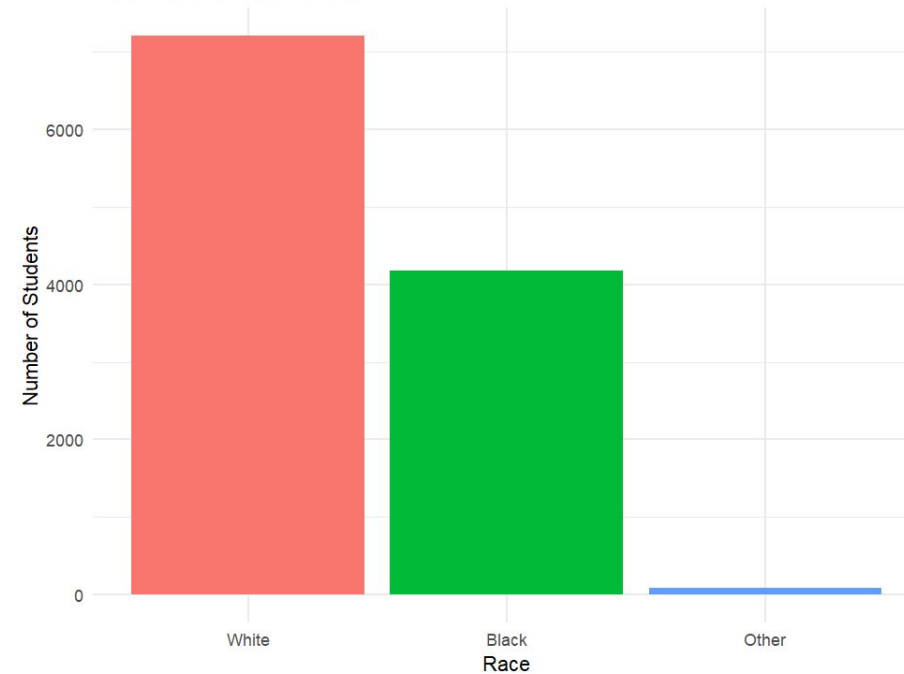
- Statistically significant outcomes here could inform education policy on intervention strategies.
- Results can help us understand if the small-class effect is **preventative** or **remediative**.
- Outcomes of the experiment could help schools manage the costs associated with smaller classrooms.
- Potential K-12-based remediation for new California legislation: AB705 / AB1705

Exploratory Data Analysis

Distribution of School Urbanicity



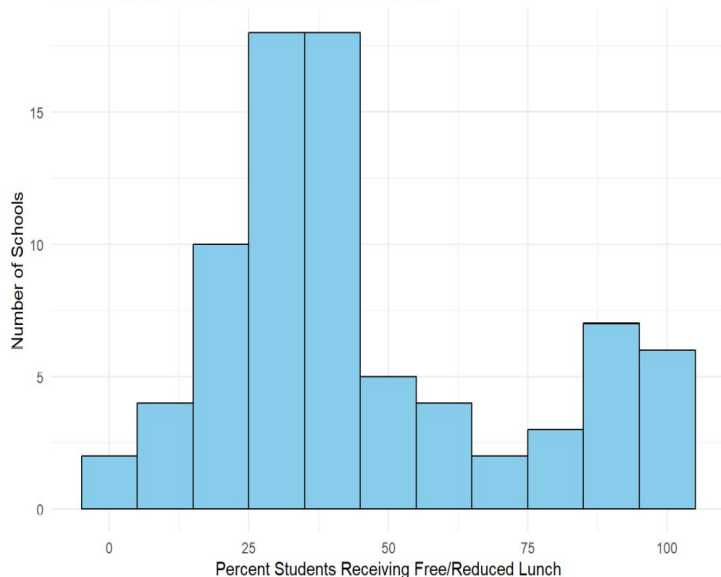
Student Race Distribution



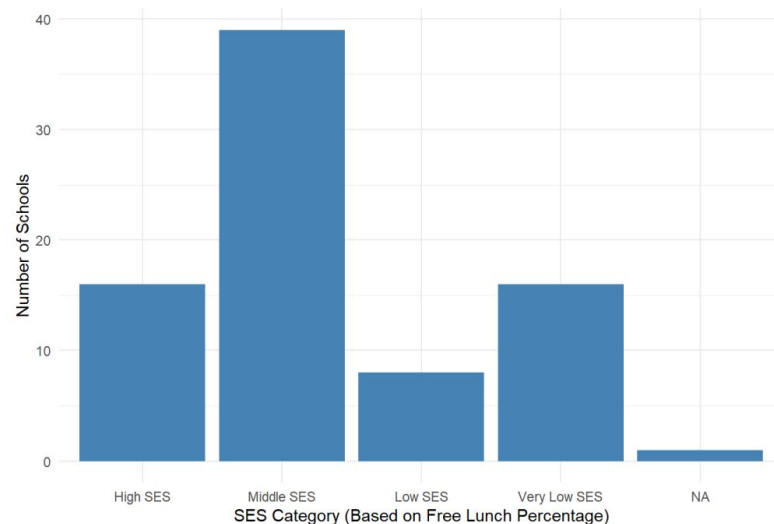
Exploratory Data Analysis

- SES = Socioeconomic Status
- measure SES using the percentage of students receiving free or reduced lunch
- higher percentage of free lunch students = lower SES

Distribution of Free/Reduced Lunch Percentage

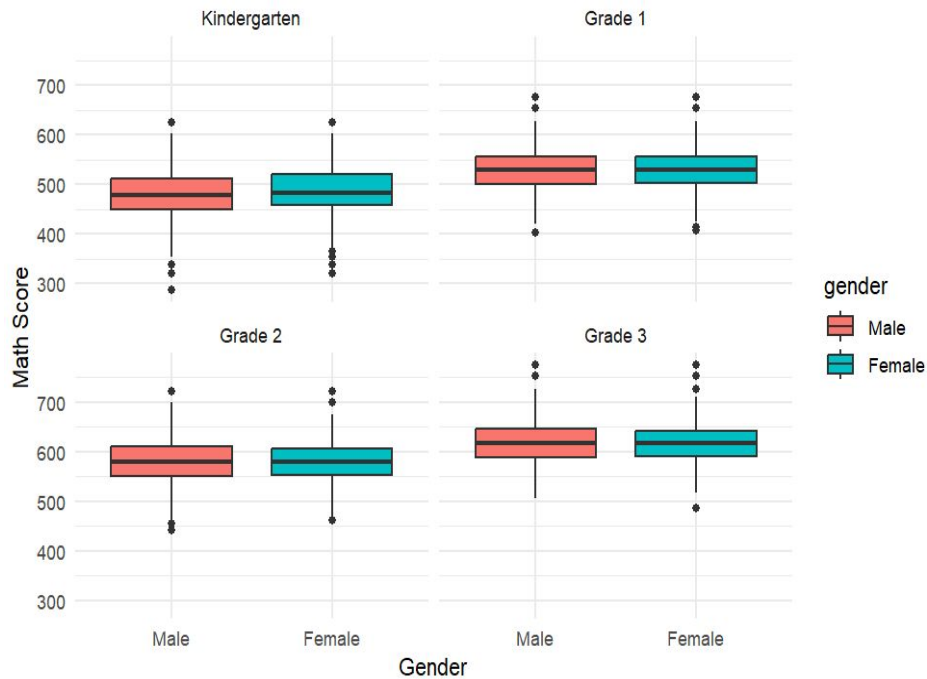


School Socioeconomic Status Distribution

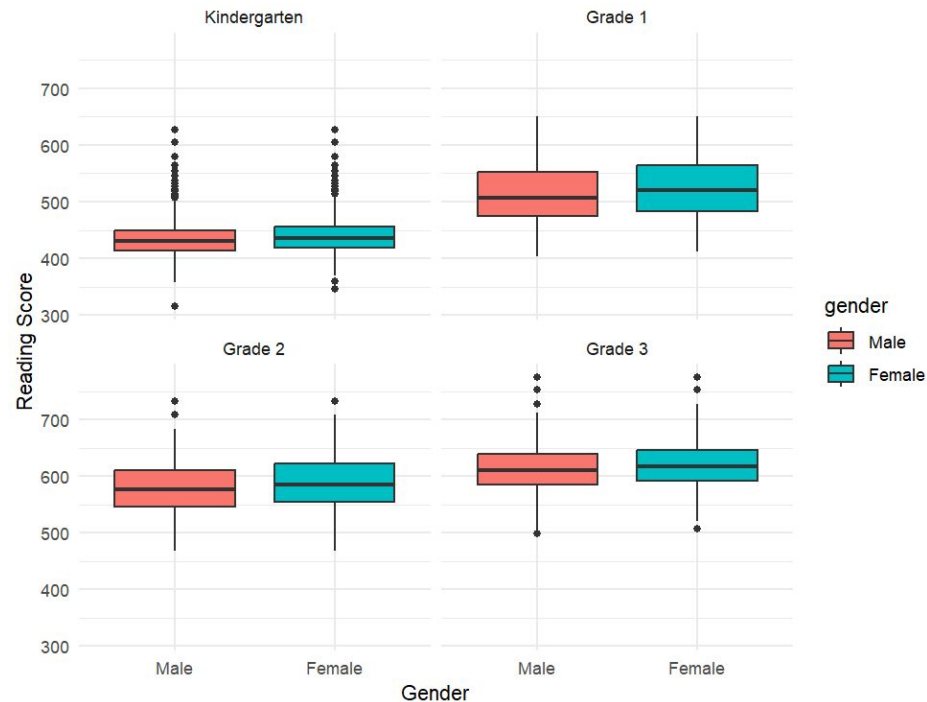


Exploratory Data Analysis

Math Scores by Gender Across Grades



Reading Scores by Gender Across Grades

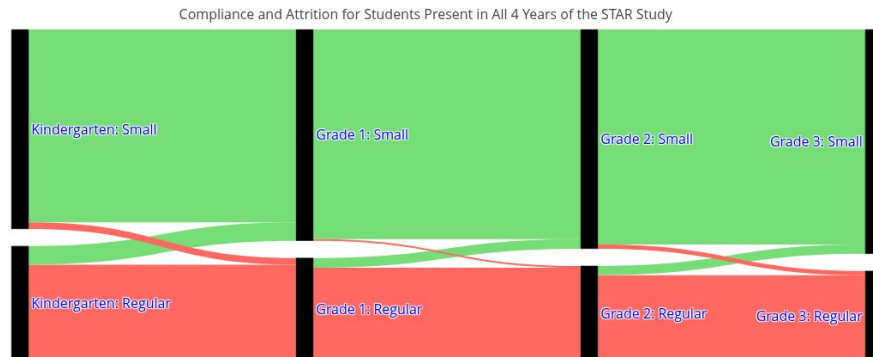


Methods: Data Wrangling

- Dataset filters
 - Students in all 4 years of study
 - Students who were not in regular class with aide
 - Race either black or white
- Trend Variable
 - Compliers: Constant regular and small class size
 - Non-compliers: Regular to small and small to regular class size switchers

Table 1: Trend Variable Sample Size post Filter

Trend	Sample Size
Constant Regular Class Size	365
Constant Small Class Size	853
Regular to Small Class Size	151
Small to Regular Class Size	38
Total	1407



Methods: Preliminary Model Fit

Model 1: Preliminary model for class size and socioeconomic status effects on early years

- Three Factor ANOVA

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + \gamma_k + \epsilon_{ijkl}$$

where $\epsilon_{ijkl} \sim N(0, \sigma^2)$

μ : overall mean

α_i : main effect of class size

β_j : main effect of free lunch

γ_k : main effect of urbanicity

Model 2: Model for class size *trend*

- Mixed Effects with random intercept for school

$$Y_{ij} = \beta_0 + \beta_1 \text{trend} + \beta_2 \text{gender} + \beta_3 \text{race} + \beta_4 \text{freelunch} + \beta_5 \text{tyears} + b_{0j} + \epsilon_{ij}$$

where $b_{0j} \sim N(0, \sigma_b^2)$, $\epsilon_{ij} \sim N(0, \sigma^2)$

- For each model, we fit candidate interactions models and will compare which provides a better fit

Methods: Model Selection

- Compare BIC Scores
- Model 1: Extra sum-of-squares F-test

$$F = \frac{(RSS_1 - RSS_2)/(df_1 - df_2)}{RSS_2/df_2}$$

Table 2: Comparison of Three-way ANOVA Candidate Models by F Test

Model	Parameters	Residual df	AIC	BIC	RSS	Sum Sq.	F-stat.	p-val.
Additive	8	1336	13990	14026	2619558			
Interaction (classsize*freelunch)	9	1335	13992	14034	2619513	44.9	0.0228	0.8799
Interaction (classsize*urbanicity)	11	1333	13996	14048	2618711	802.4	0.2040	0.8155
Interaction (classsize*freelunch*urbanicity)	18	1326	14004	14093	2608328	10382.6	0.7540	0.6261

- Model 2: Likelihood Ratio Test

Table 3: Comparison of Mixed Effects Candidate Models by Likelihood Ratio Test

Model	Parameters	AIC	BIC	-2log(L)	Chisq	p-value
Additive	10	12928	12980	12908		
Interaction (trend*freelunch)	13	12927	12994	12901	7.7188	0.0522
Interaction (trend*race)	13	12980	12999	12906	2.1038	0.5511

Final Models

- Three way ANOVA **additive** model
- Mixed Effect **additive** model

Results: Three Factor ANOVA Model

Do smaller class sizes disproportionately benefit low-income students in early elementary school?

F-Tests for group means

$$F^* = \frac{\frac{SSTR}{df\{SSTR\}}}{\frac{SSE}{df\{SSE\}}} = \frac{MSTR}{MSE}$$

$$F_{critical} = F(1 - \alpha; df_1, df_2)$$

$$H_0: \mu_i = \dots = \mu_r$$

$$H_1: \text{not all } \mu_i \text{ are the same}$$

Table 4: ANOVA Table for class size, free lunch, and urbanicity

	Df	Sum Sq	Mean Sq	F value	F_critical	Pr(>F)
Class Type	1	37239.930	37239.930	18.993	3.85	1.41e-05 ***
Free Lunch	1	95665.547	95665.547	48.790	3.85	4.47e-12 ***
School Location	3	2456.925	818.975	0.418	2.61	0.74 .
Residuals	1336	2619558.218	1960.747	NA	NA	NA

Results: Mixed Effects Model

For students who begin their academic journey in regularly sized classrooms, is it educationally advantageous to switch to smaller classes in later years?

Table 3: Results of Mixed Effects Additive Model

Variable	Estimate	Standard Error	df	t-statistic	p-value
Intercept	615.9284	3.9218	485.1381	157.051	<2e-16
trendConstantSmall	5.6360	2.3992	1263.1583	2.349	0.0190
trendRegtoSmall	3.5772	3.7306	1280.9287	0.959	0.3378
trendSmalltoReg	13.8712	7.2142	1204.8663	1.923	0.0547
genderFemale	-1.1203	1.9861	1232.0915	-0.564	0.5628
raceBlack	-13.4432	3.4631	383.7018	-3.882	1.22e-4
g3freelunchNonFree	16.5513	2.4110	1279.9178	6.865	1.04e-11
g3years	0.2121	0.1334	1171.7302	1.589	0.1123

Table 4: Results of Mixed Effects Additive Model
(Random Variable)

Group	Variance	Standard Deviation
g3schoolid (intercept)	192.2	13.85
Residual	1224.7	35.00

- Trend effect only significant for compliers
- Significant covariates
 - race
 - free lunch

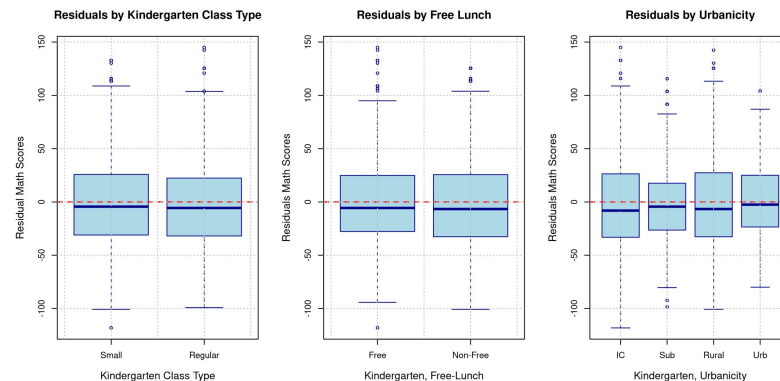
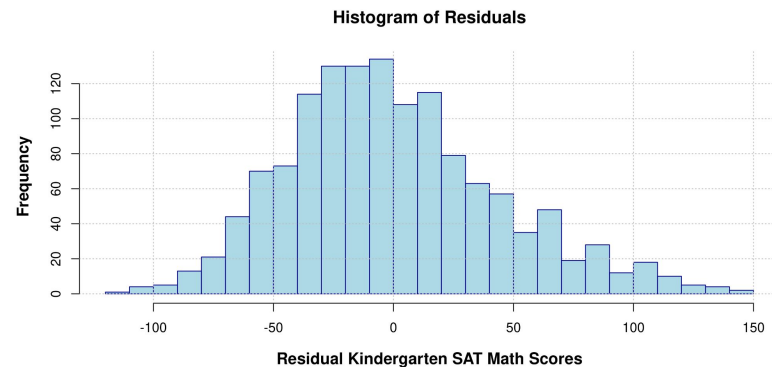
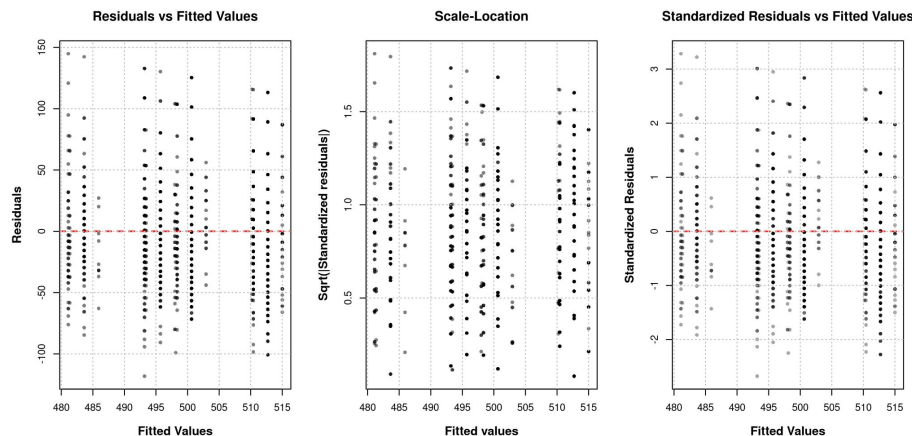
Intraclass Correlation Coefficient (ICC)

$$\rho = \frac{\sigma_a^2}{\sigma_a^2 + \sigma^2} = \frac{192.2}{192.2 + 1224.7} = 0.1356$$

- around 13.6% of variation in performance is due to school differences

Sensitivity Analysis: Three-way ANOVA

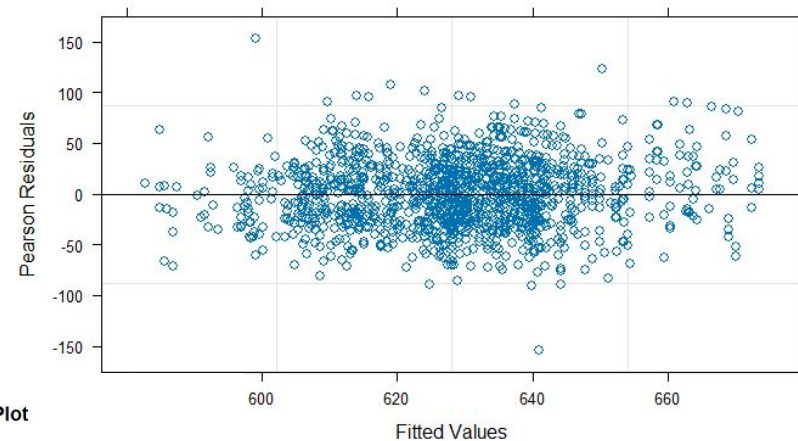
Do smaller class sizes disproportionately benefit low-income students in early elementary school?



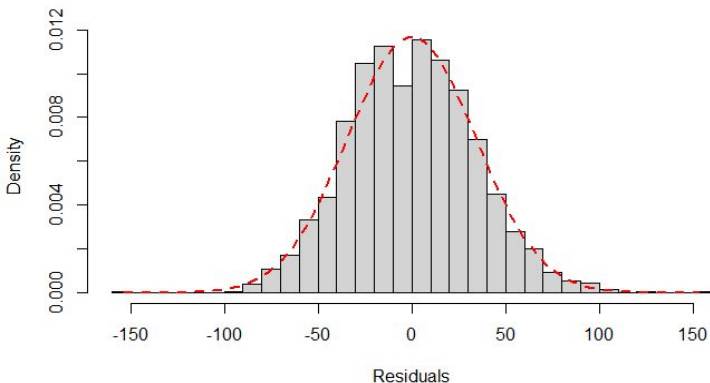
Sensitivity Analysis: Mixed Effects

For students who begin their academic journey in regularly sized classrooms, is it educationally advantageous to switch to smaller classes in later years?

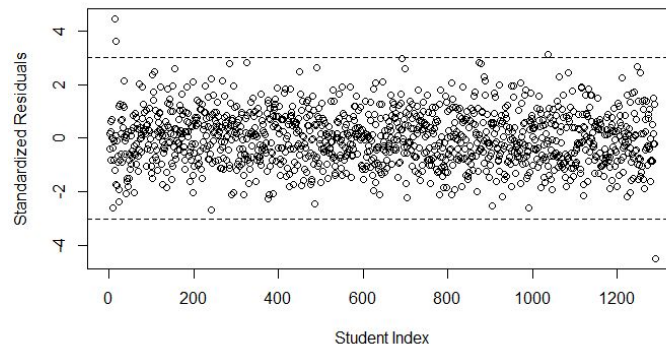
Residuals vs. Fitted



Residual Distribution



Studentized Residual Plot



Discussions

- Slight heteroscedasticity in three factor ANOVA model
 - F test should be interpreted with caution
- We used free lunch to determine socioeconomic status
- Non-compliers (students who switch from small to regular or vice-versa) assumed to be independent
 - Students may switch classes if their friend switches

Limitations

- Potential outliers in three factor ANOVA (kindergarten math scores)
- Analysis excludes regular sized classes with aide (*gkclasstype*=Ra)
- Potential confounding variables (parent involvement, school resources)
- Only math scores

Conclusions

- Class size and socioeconomic status (i.e., free lunch affiliation) group-means **do differ**.
- Data **do not suggest** that it is advantageous to switch to a smaller class in later years
 - “Not statistically significant” $\Rightarrow$ inconclusive results, not a proof.

Future Work

- Large scale, randomized, longitudinal study on class size switching.
- More modern studies with samples taken across different states.